

Ideum's Tangible Engine with Multiple Displays

Unity Engine Specific

Summary

This document explains how to get Tangible Engine working on a multi-display setup, and how the software needs to be configured in order to allow touches to work for all touch sensors involved.

Hardware Setups

Many Displays with One Touch Sensor

This is the easiest setup to get working since it only uses one touch sensor. Using the NVIDIA Quatro, open the settings and use the Mosaic/Surround feature to turn the many displays into one larger display.

Now any application will fill the entirety of the multi-displays since it thinks it's just one display. You can confirm this has worked in the Windows display settings.

Many Displays with Many Touch Sensors

This is the more complicated setup since there are many touch sensors involved. The trick here is to connect the display of one computer to the graphics card of the main computer, so only one computer is running, but has extended the other displays as if they were only a monitor.

You can check to see if this setup is working correctly by opening the Windows Display Settings and seeing that all connected displays show up with a number.

The main display (the one with the computer) should be number 1.

Software Setup

The software setup heavily depends on the hardware setup.

If you have **many displays with one touch sensor**, acting as one large display, create your software based on the chosen resolution when setting up the Mosaic/Surround configuration. Touchscript does not need to be included since the software acts like one simple application.

If you have **many displays with many touch sensors**, include Touchscript in your application so that the touches do not re-map to display 1.

- Decide your resolution based on the configuration and layout of your monitors, then rearrange them inside of the Display Settings to match the physical layout
 - One long horizontal line
 - One long vertical line
 - A square
 - Etc
- Monitors / displays should all be of the same resolution and should be mirrored in your application
 - Example) Two 4k monitors, side by side horizontally, should be the same in your application, with two 4k canvases, side by side horizontally
- Based on your final resolution, set up your scene using only display 1.
 - You can have multiple cameras in the scene, but they will all need to output to display 1.
- On each of your cameras, add a [Touchscript](#) Standard Layer
 - Set your hit test options based on your application's functionality
- Change your Event System to use the Touch Script Input Module and add the Standard Input script
- Add the Tangible Engine prefab to the scene
- For testing and in editor, place a Simulator View prefab into the scene and set it's Canvas Scaler reference resolution to the correct resolution
 - This makes sure that the tangible overlay positions the simulated tangibles correctly based off of your mouse / touch points
- Build your application as you normally would
- Run the application through the command line or batch script and force it to open in the correct resolution
 - Ex) MyApp.exe -screen-width 7860 -screen-height 2160 -popupwindow -screen-fullscreen 0
 - This will keep the application from trying to open fullscreen on only one display, and will instead "overflow" onto the other displays